

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2075

Dewing 138 kV
October 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2075. The Interconnection Customer proposes a 273.4 MW battery energy storage facility interconnecting at the Dewing 138 kV substation in the Pennsylvania area. The request was received on April 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 605 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$111,117,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Knott-South Springfield 138kV is overloaded to 177.6% of its 220 MVA emergency rating for the loss of the Allegheny Hydro No 9-Kirkland Park 138kV. The project contributes 13.0% DFAX (35.6 MW) to this overload.

The Dewing-Handsoma Lake 138kV is overloaded to 160.7% of its 180 MVA emergency rating for the loss of the Brodhead S-Marion N Substation 138kV Line. The project contributes 68.9% DFAX (188.2 MW) to this overload.

For the loss of Dewing-Handsoma Lake 138kV, the Dewing-Harrison St 138kV Line loads to 148.3% of its 187 MVA rating. Project DFAX: 100.0%.

Harrison St-St. Charles Power Plant 138kV L: 142.0% loading (198 MVA) following the outage of the Handsoma Lake-Dewing 138kV; DFAX = 100.0%.

The Knott-Primm 138kV is overloaded to 141.2% of its 289 MVA emergency rating for the loss of the Kirkland Park-Allegheny Hydro No 9 138kV. The project contributes 8.5% DFAX (23.2 MW) to this overload.

Johns Creek Substation-Elizabethton 138kV: 139.4% loading (228 MVA) following the outage of the Newark-Phoenix Hydro Project Substation 138kV; DFAX = 5.2%.

The Elizabethton-Tesoro Kenai Cogeneration Plant 138kV is overloaded to 136.4% of its 236 MVA emergency rating for the loss of the Phoenix Hydro Project-Newark 138kV line. The project contributes 5.2% DFAX (14.2 MW) to this overload.

The Rock Tavern-St. Charles Power Plant 138kV is overloaded to 134.5% of its 212 MVA emergency rating for the loss of the Dewing-Handsoma Lake 138kV L. The project contributes 100.0% DFAX (273.4 MW) to this overload.

Kirkland Park-Allegheny Hydro No 9 138kV: 132.4% loading (278 MVA) following the outage of the Knott-South Springfield 138kV L; DFAX = 7.0%.

For the loss of Lions Road-Piney Crk Project 500kV, the Johns Crk-Primm 138kV Line loads to 131.1% of its 337 MVA rating. Project DFAX: 6.2%.

The El Paso 500/345kV Transformer is overloaded to 117.6% of its 864 MVA emergency rating for the loss of the Ganso-El Paso 500kV. The project contributes 7.9% DFAX (21.6 MW) to this overload.

Skiland 500/345 kV Autotransformer: 115.0% loading (781 MVA) following the outage of the Ganso-Handsome Lake 500kV; DFAX = 8.8%.

The Lions Road 345-138 kV Auto is overloaded to 114.5% of its 381 MVA emergency rating for the loss of the Tyhee Substation-Stent Tap 138kV. The project contributes 5.1% DFAX (14.0 MW) to this overload.

The Greg Stblla Vista-Weterlo 138kV is overloaded to 113.3% of its 172 MVA emergency rating for the loss of the Dewing-Handsome Lake 138kV. The project contributes 37.7% DFAX (103.2 MW) to this overload.

For the loss of Handsome Lake-Piney Crk Project 500kV, the Handsome Lake 500/345kV Autotransformer loads to 107.9% of its 724 MVA rating. Project DFAX: 28.3%.

The Gilstrap-Ganso 345kV is overloaded to 104.7% of its 1,041 MVA emergency rating for the loss of the Ganso-El Paso 500kV. The project contributes 12.9% DFAX (35.4 MW) to this overload.

The Chamberlin-Imperial 230kV is overloaded to 101.6% of its 1,691 MVA emergency rating for the loss of the Hylebos-Ravenna 500kV. The project contributes 5.8% DFAX (16.0 MW) to this overload.

The Gilstrap-ElPaso Substation 345kV is overloaded to 101.5% of its 1,175 MVA emergency rating for the loss of the Ganso Substation-El Paso 500kV. The project contributes 14.5% DFAX (39.7 MW) to this overload.

The Junction City-Greg Stbella Vista 138kV is overloaded to 101.4% of its 189 MVA emergency rating for the loss of the Dewing-Handsome Lake 138kV. The project contributes 37.7% DFAX (103.2 MW) to this overload.

The Old Forge-Ravenna 230kV is overloaded to 100.6% of its 1,620 MVA emergency rating for the loss of the Ravenna-Hylebos 500kV. The project contributes 6.4% DFAX (17.6 MW) to this overload.

Navarre-Underwood 138kV Line: 100.0% loading (170 MVA) following the outage of the Hndsome Lake 345/138kV Transformer Bank; DFAX = 19.1%.

Skiland-Corning 345kV: 100.0% loading (961 MVA) following the outage of the Ganso Substation-Handsome Lake 500kV; DFAX = 9.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Knott-South Springfield 138kV	Rebuild line with higher-capacity conductor	\$3,983,000	84% (\$3,331,000)
Dewing-Handsome Lake 138kV	Rebuild line with higher-capacity conductor	\$6,486,000	100% (\$6,486,000)
Harrison St-Dewing 138kV	Rebuild line with higher-capacity conductor	\$7,738,000	100% (\$7,738,000)
Harrison St-St. Charles Power Plant 138kV L	Rebuild line with higher-capacity conductor	\$15,270,000	100% (\$15,270,000)
Knott-Primm 138kV	Rebuild line with higher-capacity conductor	\$2,326,000	76% (\$1,764,000)
Johns Creek Substation-Elizabethton 138kV	Rebuild line with higher-capacity conductor	\$2,024,000	73% (\$1,480,000)
Elizabethton-Tesoro Kenai Cogeneration Plant 138kV	Rebuild line with higher-capacity conductor	\$3,491,000	73% (\$2,553,000)
Rock Tavern-St. Charles Power Plant 138kV	Rebuild line with higher-capacity conductor	\$4,523,000	100% (\$4,523,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kirkland Park-Allegheny Hydro No 9 138kV	Rebuild line with higher-capacity conductor	\$13,120,000	100% (\$13,120,000)
Johns Crk-Primm Substation 138kV L	Rebuild line with higher-capacity conductor	\$8,990,000	74% (\$6,618,000)
El Paso 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$37,746,000	8% (\$3,035,000)
Skiland 500/345 kV Autotransformer	Replace transformer with higher-rated unit	\$27,295,000	20% (\$5,424,000)
Lions Road 345-138 kV Auto	Replace transformer with higher-rated unit	\$16,119,000	72% (\$11,642,000)
Westerlo-Greg Stbella Vista 138kV	Rebuild line with higher-capacity conductor	\$3,112,000	100% (\$3,112,000)
Handsome Lake 500/345kV Autotransformer	Replace transformer with higher-rated unit	\$16,791,000	92% (\$15,442,000)
Gilstrap-Ganso 345kV	Reconductor line	\$13,156,000	14% (\$1,819,000)
Imperial-Chamberlin 230kV	Reconductor line	\$2,699,000	6% (\$149,000)
Gilstrap-ElPaso Substation 345kV	Reconductor line	\$30,553,000	9% (\$2,641,000)
Greg Stbella Vista-Jct City 138kV	Reconductor line	\$798,000	100% (\$798,000)
Raenna-Old Forge 230kV L	Reconductor line	\$6,990,000	8% (\$592,000)
Navarre Substation-Underwood 138kV Line	Reconductor line	\$2,606,000	54% (\$1,399,000)
Skiland-Corning 345kV	Reconductor line	\$9,809,000	22% (\$2,181,000)
	Total allocated to project		\$111,117,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.